

SAFE STUDENT ASSISTANT AGENT

Topic 07 — Submission Test Evidence

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Course: AI Engineering

1. Test Case — Course Search

- **Purpose:** To verify that the agent can search for a course and return the correct information.
- **Screenshot:**

```
PS D:\Code\AI\homework\safe-student-agent-homework> python main.py
=====
SAFE STUDENT ASSISTANT AGENT
=====

Enter your role (student/admin): student
Enter the request: Find Python Programming

=====
AI RESPONSE
=====

User: Find Python Programming
Role: student
-----
Route: search

ReAct Step: 1/5
Action: search_course
Arguments: {'keyword': 'Python Programming'}
Observation: {'success': True, 'courses': [{'id': 1, 'name': 'Python Programming',
'teacher': 'Mr. Dara', 'schedule': 'Monday 6:00 PM - 8:00 PM'}]}

ReAct Step: 2/5
Final Answer:
The course "Python Programming" with the ID 1 is available. Would you like to regis
ter for this course, check the course schedule, or do something else?
PS D:\Code\AI\homework\safe-student-agent-homework> 
```

2. Test Case — Permission Control

- **Purpose:** To verify that students are prevented from using the restricted registration tool.
- **Screenshot:**

```
PS D:\Code\AI\homework\safe-student-agent-homework> python main.py
=====
SAFE STUDENT ASSISTANT AGENT
=====

Enter your role (student/admin): student
Enter the request: Register student 1001 for Data Analytics

=====
AI RESPONSE
=====

User: Register student 1001 for Data Analytics
Role: student
-----
Route: register
Request Data:
{'student_id': 1001, 'course_keyword': 'Data Analytics'}
Observation: {'success': False, 'error_code': 'PERMISSION_DENIED', 'error': "Permission denied: role 'student' cannot use 'register_course'."}
Final Answer:
I could not register the student because registration requires admin permission.
PS D:\Code\AI\homework\safe-student-agent-homework> |
```

3. Test Case — Admin Registration + Human-in-the-Loop

- **Purpose:** To verify the multi-step registration workflow and human approval for a high-risk action.
- **Screenshot:**

```
PS D:\Code\AI\homework\safe-student-agent-homework> python main.py
=====
SAFE STUDENT ASSISTANT AGENT
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Enter your role (student/admin): admin
Enter the request: Register student 1002 for Korean Language

=====
AI RESPONSE
=====

User: Register student 1002 for Korean Language
Role: admin
-----
Route: register
Registration workflow detected.
Action: search_course
Arguments: {'keyword': 'Korean Language'}
Observation: {'success': True, 'courses': [{'id': 4, 'name': 'Korean Language', 'teacher': 'Ms. Mina', 'schedule': 'Friday 6:00 PM - 8:00 PM'}]}

Course found: Korean Language (ID: 4)
Action: register_course
Arguments: {'student_id': 1002, 'course_id': 4}

=====
HUMAN APPROVAL REQUIRED
=====

Action: Register student 1002 for Korean Language
Risk Level: HIGH
=====

Do you want to approve this action? [y/N]: y
Observation: {'success': True, 'message': 'Student 1002 successfully registered for Korean Language.', 'course': {'id': 4, 'name': 'Korean Language', 'teacher': 'Ms. Mina', 'schedule': 'Friday 6:00 PM - 8:00 PM'}}
Final Answer:
Student 1002 successfully registered for Korean Language.
PS D:\Code\AI\homework\safe-student-agent-homework> |
```